

d & f-BLOCK ELEMENTS

- 1** Transition element is defined as one which has–
 - (1) Fully filled $(n - 1)$ d subshell
 - (2) Incompletely filled nd subshell in ground or in higher oxidation state
 - (3) Incompletely filled $(n - 1)$ d subshell in ground state
 - (4) Incompletely filled $(n - 1)$ d subshell in ground state or any one of its oxidation state
- 2** Zn, Cd and Hg are not regarded as transition elements on the basis of electronic configuration because–
 - (1) they do not have electrons in d-subshell
 - (2) their d and s-subshells remain fully filled in ground and higher oxidation states
 - (3) their s-subshell remain incomplete in ground and ionic state
 - (4) their penultimate shell remains completely filled in ground and in higher oxidation state
- 3** In general melting point of transition elements in a series can be explained by following relation–
 - (1) Melting point $\propto$ atomic weight
 - (2) Melting point $\propto$ no of unpaired valence electrons
 - (3) Melting point $\propto$ Internuclear distance.
 - (4) Melting point $\propto \frac{1}{\text{atomic weight}}$
- 4** In the series Sc to Zn the enthalpy of atomisation of 'Zn' is the lowest because–
 - (1) Zn exist in molecular state
 - (2) In the crystal of Zn interatomic distance is small
 - (3) IP of Zn is minimum in the series
 - (4) No 3d orbital involved in metallic bonding.
- 5** Which among the following orders of second IP is correct?
 - (1) $\text{Mn} > \text{Cr}$
 - (2) $\text{Zn} > \text{Cu}$
 - (3) $\text{Na} > \text{Cu}$
 - (4) $\text{Cr} > \text{K}$
- 6** Which among the following statement regarding IP is incorrect?
 - (1) IP_1 of Mn is greater than its IP_2
 - (2) IP of Mn^{2+} is greater than IP of Fe^{2+}
 - (3) IP_3 of Mn is greater than IP_3 of Cr
 - (4) IP of 3d series element increases from Sc to Zn in irregular manner.
- 7** Most common oxidation states are matched below with the elements Which one is mismatched ?
 - (1) Iron (+2, +3)
 - (2) Chromium (+1, +2)
 - (3) Manganese (+2, +7)
 - (4) Titanium (+3, +4)
- 8** Which among the following statements is incorrect
 - (1) In d-block elements oxidation state differ by unity.
 - (2) In p-block metals oxidation state differ by two units.
 - (3) In a group of p-block lower oxidation states are favoured by the heavier members.
 - (4) In a group of d-block higher oxidation states are favoured by the lighter member.
- 9** Which among the following order of oxidising character is correct?
 - (1) $\text{CrO}_3 > \text{MoO}_3$
 - (2) $\text{K}_2\text{Cr}_2\text{O}_7 > \text{KMnO}_4$
 - (3) $\text{Fe}(\text{CO})_5 > \text{Mn}(\text{CO})_5$
 - (4) $\text{V}_2\text{O}_3 > \text{V}_2\text{O}_5$
- 10** Which of the following transition element does not exhibit variable oxidation state?
 - (1) Mercury
 - (2) Scandium
 - (3) Copper
 - (4) All of the above
- 11** Which of the following configurations of 3d series metals exhibits the largest number of oxidation states?
 - (1) $[\text{Ar}] 3d^8 4s^2$
 - (2) $[\text{Ar}] 3d^{10} 4s^1$
 - (3) $[\text{Ar}] 3d^5 4s^2$
 - (4) $[\text{Ar}] 3d^7 4s^2$
- 12** Cr^{2+} is reducing in nature because–
 - (1) In Cr^{2+} , configuration changes from d^4 to d^3 to achieve half filled t_{2g} .
 - (2) Cr^{2+} gain an electron to achieve d^5 configuration
 - (3) Cr^{2+} give an electron to achieve d^5 configuration
 - (4) In Cr^{2+} configuration changes from d^4 to d^3 to achieve half filled d-subshell.

- 13** The ability of fluorine to stabilize the highest oxidation states in CrF_6 , VF_5 is due to—
 (1) Oxidising property of higher oxidation states
 (2) higher lattice energy
 (3) High bond enthalpy due to covalent character
 (4) Ability to form multiple bond.
- 14** Highest stable Mn fluoride is MnF_4 where as highest Mn oxide is Mn_2O_7 due to—
 (1) O atom is smaller than F.
 (2) Oxygen has ability to form multiple bonds
 (3) Mn^{7+} does not exist
 (4) F can not stabilise higher oxidation states.
- 15** Which among the following order of oxidising power is correct?
 (1) $\text{CrO}_4^- < \text{MoO}_4^- < \text{WO}_4^-$
 (2) $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$
 (3) $\text{CrO}_3 < \text{MnO}_2 < \text{Fe}_2\text{O}_3$
 (4) $\text{Pb}^{4+} < \text{Sn}^{4+} < \text{C}^{4+}$
- 16** Which of the following compound decomposes on heating to give metal?
 (1) KNO_3 (2) CaCO_3
 (3) ZnCO_3 (4) AgNO_3
- 17** Which of the following pair of ions have same value of "spin-only" magnetic moment?
 (1) Cu^+ , Cu^{2+} (2) Co^{3+} , Fe^{2+}
 (3) Ti^{2+} , V^{2+} (4) Sc^{2+} , Zn^{+2}
- 18** The magnetic moment of divalent ion in aqueous solution if its atomic number is 25 is —
 (1) 4.9 B.M (2) 5.92 B.M
 (3) 3.8 B.M (4) 2.8 B.M
- 19** In which of the following metal ion frequency of light absorbed lies in visible region?
 (1) Sc^{3+} (2) Cu^+
 (3) Cd^{2+} (4) Cr^{3+}
- 20** Which among the following pair is of interstitial compounds ?
 (1) SiH_4 , LiH
 (2) Fe_3H , Mn_4N
 (3) Al_4C_3 , SiC
 (4) TiCl_4 , MnO_2
- 21** Which of the following statement regarding interstitial compounds is incorrect ?
 (1) They are chemically inert
 (2) They are soft and nonconductive
 (3) They retain metallic conductivity
 (4) They have high melting point.
- 22** Which among the following alloy does not contain copper?
 (1) Brass (2) Bronze
 (3) German Silver (4) Type metal
- 23** Alloys are formed by atoms with metallic radii that are within about—
 (1) 15 percent of each other
 (2) 50 percent of each other
 (3) 30 percent of each other
 (4) 75 percent of each other
- 24** Cu^+ ion is not stable in aqueous solution, due to—
 (1) d^{10} configuration
 (2) disproportionation
 (3) less charge
 (4) pseudo inert gas configuration
- 25** Atomic radii of Zr (160 pm) and Hf (159 pm) is a consequence of—
 (1) Transition contraction
 (2) Inert pair effect
 (3) poor shielding of 4f electrons
 (4) Actinoid contraction